

### Challenge (Habanero)

**Q1.** Solve for  $x$  in  $2x + 5 = 11$ .

- (A)  $x = 2$
- (B)  $x = 3$
- (C)  $x = 4$
- (D)  $x = 5$
- (E)  $x = 6$

**Q2.** A recipe for making cookies requires  $\frac{3}{4}$  cup of sugar. If you want to make  $2 \cdot \frac{3}{4}$  cups of cookies, how much sugar will you need in total, where  $s$  is the amount of sugar needed?

- (A)  $s = 1 \cdot \frac{1}{2}$
- (B)  $s = 1 \cdot \frac{1}{4}$
- (C)  $s = 1 \cdot \frac{3}{2}$
- (D)  $s = 2$
- (E)  $s = 1 \cdot \frac{3}{4}$

**Q3.** Rearrange the formula  $A = \frac{1}{2}bh$  to isolate  $h$ .

- (A)  $h = \frac{1}{2} \cdot \frac{A}{b}$
- (B)  $h = 2 \cdot \frac{A}{b}$
- (C)  $h = \frac{A}{2b}$
- (D)  $h = 2Ab$
- (E)  $h = \frac{2A}{b}$

**Q4.** In a construction project, the cost  $C$  of building a house is given by  $C = 2500 + 300n$ , where  $n$  is the number of rooms. Solve for  $n$  when  $C = 4300$ .

- (A)  $n = 5$
- (B)  $n = 6$
- (C)  $n = 7$
- (D)  $n = 8$
- (E)  $n = 9$

**Q5.** Solve for  $x$  in  $\frac{2x}{5} = 3$ .

- (A)  $x = 5$
- (B)  $x = 6$
- (C)  $x = 7.5$
- (D)  $x = 8$
- (E)  $x = 9$

**Q6.** Rearrange the formula  $V = \frac{1}{3}\pi r^2 h$  to isolate  $r$ .

- (A)  $r = \sqrt{\frac{3V}{\pi h}}$
- (B)  $r = \sqrt{\frac{\pi h}{3V}}$
- (C)  $r = \frac{3V}{\pi h}$
- (D)  $r = \frac{\pi h}{3V}$
- (E)  $r = \sqrt{\frac{V}{3\pi h}}$

**Q7.** A chef needs to package  $4\frac{1}{2}$  pounds of cake into boxes that hold  $\frac{3}{4}$  pound each. If  $x$  represents the number of boxes, solve for  $x$ .

- (A)  $x = 5$
- (B)  $x = 6$
- (C)  $x = 7$
- (D)  $x = 8$
- (E)  $x = 9$

**Q8.** Solve for  $y$  in  $2y - 4 = 12$ .

- (A)  $y = 5$
- (B)  $y = 6$
- (C)  $y = 7$
- (D)  $y = 8$
- (E)  $y = 9$

#### Open Ended Questions (FRQ)

**Q9.** Solve the equation  $x + 2 = 7$  for  $x$ . Then, use the value of  $x$  to find the area of a triangle with base  $x$  and height 4. Show your work.

**Q10.** The length  $L$  of a rectangular garden is given by  $L = 2w + 5$ , where  $w$  is the width. Find the value of  $w$  when  $L = 17$ . Then, find the area of the garden using the formula  $A = L \cdot w$ . Show your work.

#### Chef's Tips

- Remember to check your work
- Use the given formulas to solve for the unknown variable
- Show all work and explain your reasoning